C PROGRAMMING — ASSIGNMENT 1

Task 7: Control Type Conversion and Precision

Marks: 4 | Total assignment: 50 marks

Conversion points

Integer promotions can convert short and char values to int before arithmetic. Usual arithmetic conversions determine a common type for mixed expressions. Integer division truncates toward zero: 5/2 is 2, not 2.5. Casting a large long to int can lose information or produce an implementation-defined result if the value is not representable. Convert money to paise using a documented rounding rule; for example, round a non-negative rupee amount to the nearest paise before converting to an integer.

Three surprises

1) 5 / 2 prints 2 because both operands are integers. Use 5.0 / 2 for 2.5. 2) (int)3.99 prints 3 because conversion truncates toward zero, it does not round. 3) 0.1 + 0.2 == 0.3 is commonly false with binary double because these decimal fractions are not exactly representable. Another common trap is 1 / 100 yielding 0. Avoid it for charge calculations by using integer paise or floating operands with explicit rounding.

Error experiment

The program below compares repeated float addition against integer-paise accumulation. Record the printed result on your system. Exact integer paise accumulation is reliable while the total remains within the chosen integer range; float accumulation is approximate and its error depends on magnitude, order and number of additions.

C Source Code / Implementation

```
#include <stdio.h>
#include <math.h>
int main(void) {
    printf("5 / 2 = %d\n", 5 / 2);
    printf("(int)3.99 = %d\n", (int)3.99);
    printf("0.1 + 0.2 == 0.3 ? %s\n", (0.1 + 0.2 == 0.3) ? "true" : "false");
    printf("1 / 100 = %d; 1.0 / 100 = %.4f\n", 1 / 100, 1.0 / 100);
    float rupees = 0.0f;
    long long paise = 0;
    for (int i = 0; i < 4000; ++i) {
        rupees += 0.1f;
        paise += 10;
    }
    printf("4000 x Rs 0.10: float=%.7f, integer paise=%lld (Rs %lld.%02lld)\n",
           rupees, paise, paise / 100, paise % 100);
    return 0;
}
```

Submission note

Compile and run the code on your own system, save actual terminal output/screenshots, and include them in your report. Review the code and be prepared to explain it during the viva. Any machine-specific measurements must come from your own execution.